

1 Solution — GIAM 7.2

Problem 4

1.1 Problem

Prove that, if n is odd, any $n \times n$ chessboard with a square the same color as one of its corners pruned can be tiled by dominoes.

1.2 Setup

Index the squares by (i, j) , $i, j \in \{1, 2, \dots, n\}$, and color each square according to the parity of $i + j$:

$$\text{color}(i, j) = \begin{cases} \text{dark} & \text{if } i + j \text{ is even,} \\ \text{light} & \text{if } i + j \text{ is odd.} \end{cases}$$

The four corners are at $(1, 1)$, $(1, n)$, $(n, 1)$, (n, n) ; their $i + j$ values are 2 , $n + 1$, $n + 1$, $2n$. For **odd** n , *all four* of these are even, so **every corner is dark**. (For even n , only the two diagonally-opposite pairs share a color — the situation of Problem 3.)

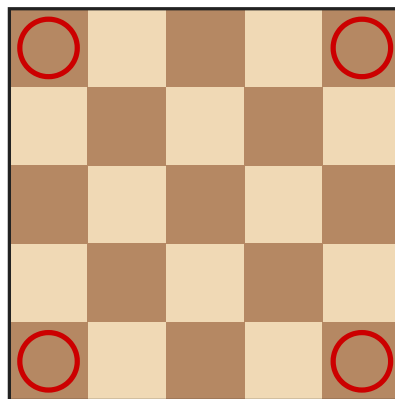


Figure 1.1: A 5×5 chessboard with all four corners circled in red. When n is odd, every corner has $i+j$ even, so all corners share the ‘dark’ color.

1.3 Counts

For odd n , the dark squares sit at the (i, j) with $i + j$ even — equivalently, i and j have the same parity. The numbers of odd and even values in $\{1, \dots, n\}$ are $(n + 1)/2$ and $(n - 1)/2$ respectively, so

$$|\text{dark}| = \left(\frac{n+1}{2}\right)^2 + \left(\frac{n-1}{2}\right)^2 = \frac{n^2 + 1}{2}, \quad |\text{light}| = n^2 - |\text{dark}| = \frac{n^2 - 1}{2}.$$

So the full board has *one extra dark square*. Removing one dark square $S = (r, c)$ — the “same color as a corner” — leaves

$$|\text{dark after}| = \frac{n^2 - 1}{2}, \quad |\text{light after}| = \frac{n^2 - 1}{2}.$$

The counts are balanced, which is necessary (Problem 3) but not by itself sufficient for a tiling. We need an *explicit construction*.

1.4 Construction

Write $S = (r, c)$ with $r + c$ even. Since the sum is even, r and c have the *same parity*. Two cases.

1.4.1 Case I — r and c both odd

Decompose the board into three horizontal strips meeting along the row of S :

1. **Top strip** — rows $1, 2, \dots, r - 1$. There are $r - 1$ rows (*even*, since r is odd). Pair adjacent rows: $\{1, 2\}, \{3, 4\}, \dots, \{r - 2, r - 1\}$. Each pair is a $2 \times n$ rectangle, tileable with n vertical dominoes.
2. **Row r minus S** — this row splits at column c into two pieces:
 - left piece (columns $1, \dots, c - 1$): length $c - 1$, *even* since c is odd;
 - right piece (columns $c + 1, \dots, n$): length $n - c$, *even* since n, c both odd. Tile each piece with horizontal dominoes — $(c - 1)/2$ on the left, $(n - c)/2$ on the right.
3. **Bottom strip** — rows $r + 1, \dots, n$. There are $n - r$ rows (*even*, since n, r both odd). Pair adjacent rows as in step 1.

Every cell of the board (except S) is covered exactly once. ✓

1.4.2 Case II — r and c both even

For Case II the row decomposition fails ($r - 1$ is now *odd*, so rows above r cannot be paired into $2 \times n$ strips). Instead, use a *small block* trick:

1. **Central 3×3 block** — at rows $r - 1, r, r + 1$ and columns $c - 1, c, c + 1$, with $S = (r, c)$ as its center. Since r is even and $2 \leq r \leq n - 1$ (any even index $r \in \{2, 4, \dots, n - 1\}$ satisfies $2 \leq r \leq n - 1$ automatically, since $n - 1$ is even for n odd), and similarly for c , this block fits inside the board. Tile the block-minus- S as follows:

- $(r - 1, c - 1)$ - $(r - 1, c)$ horizontal,
- $(r - 1, c + 1)$ - $(r, c + 1)$ vertical,
- $(r, c - 1)$ - $(r + 1, c - 1)$ vertical,
- $(r + 1, c)$ - $(r + 1, c + 1)$ horizontal.

Four dominoes covering all eight cells of the block other than S .

2. **Top strip** — rows $1, \dots, r - 2$, all n columns. This is an $(r - 2) \times n$ rectangle. Since r is even, $r - 2$ is *even*; tile each column with $(r - 2)/2$ vertical dominoes.
3. **Bottom strip** — rows $r + 2, \dots, n$, all n columns. Size $(n - r - 1) \times n$. Since n is odd and r is even, $n - r - 1$ is *even*; tile each column with vertical dominoes.
4. **Left strip** — rows $r - 1, r, r + 1$, columns $1, \dots, c - 2$. Size $3 \times (c - 2)$. Since c is even, $c - 2$ is *even*; tile each row with $(c - 2)/2$ horizontal dominoes.
5. **Right strip** — rows $r - 1, r, r + 1$, columns $c + 2, \dots, n$. Size $3 \times (n - c - 1)$. Since $n - c - 1$ is *even*, tile each row with horizontal dominoes.

The five pieces partition the board minus S :

$$\text{cells covered} = \underbrace{8}_{\text{block}-S} + \underbrace{(r - 2)n}_{\text{top}} + \underbrace{(n - r - 1)n}_{\text{bottom}} + \underbrace{3(c - 2)}_{\text{left}} + \underbrace{3(n - c - 1)}_{\text{right}} = n^2 - 1,$$

after simplification. ✓

1.4.3 Why The Two Cases Suffice

Because $r + c$ is even, r and c must have the same parity. Case I (both odd) and Case II (both even) together exhaust the possibilities. Therefore an explicit tiling exists for *every* allowed S . ■

1.5 Two Worked Examples on a 5×5 Board

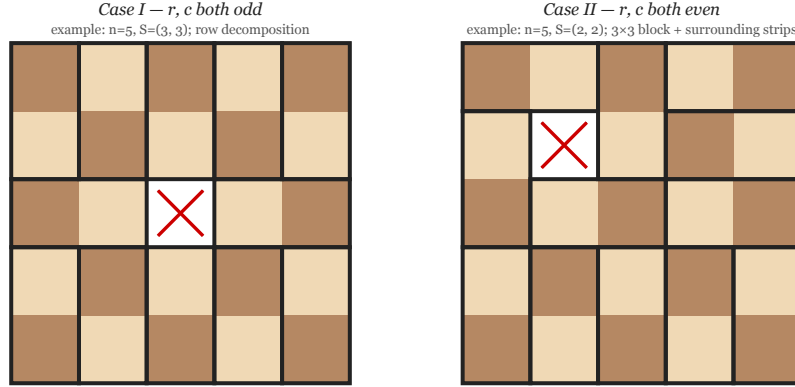


Figure 1.2: Left (Case I): $n=5, S=(3,3)$; top and bottom row-pairs are tiled vertically, and row 3 splits into two horizontal-domino segments around the removed center. Right (Case II): $n=5, S=(2,2)$; the 3×3 block at rows 1-3, cols 1-3 (minus S) carries four dominoes, with the right 3×2 strip and the bottom 2×5 strip filling the remaining cells.

For $n = 5$:

- **Case I** ($S = (3, 3)$, the center): $5 + 2 + 5 = 12$ dominoes. The two top rows are five vertical dominoes, row 3 is two horizontal dominoes spanning columns 1-2 and 4-5, and the bottom two rows are again five vertical dominoes.
- **Case II** ($S = (2, 2)$): $4 + 3 + 5 = 12$ dominoes. The 3×3 block in the top-left supplies four dominoes around the removed center, the right 3×2 strip supplies three horizontal dominoes, and the bottom 2×5 strip supplies five vertical dominoes.

1.6 Edge Case — $n = 1$

A 1×1 board has a single square — necessarily a corner — and that square is dark. “Removing the same color as a corner” *removes* the only square, leaving the empty board. The empty board is vacuously tiled by 0 dominoes. So the claim holds trivially at $n = 1$. (The construction above assumes $n \geq 3$ implicitly.)

1.7 Remarks

- **Why the parity of n matters.** When n is *even*, the diagonally-opposite corners share a color but the *adjacent* corners do not, so the four corners come in two color pairs. This is exactly Problem 3 — and removing *one* corner-color square leaves an unbalanced count, just like the mutilated chessboard. The odd- n case is opposite in flavor: the imbalance is built into the board itself (one extra dark square), and removing a dark square *restores* balance.
- **Gomory’s theorem.** For an *even*-side $n \times n$ board with the natural coloring, removing any *one* dark and any *one* light square leaves a tileable region. The proof uses a Hamiltonian cycle through the cells; in fact, the result of this problem can also be proved via a Hamiltonian-cycle / Hamiltonian-path argument on a suitably modified board, although the explicit “row + small block” construction above is much more elementary.
- **Two cases, one symmetry.** The split into Case I (odd-odd) and Case II (even-even) is not arbitrary: the $n \times n$ grid graph has a $\mathbb{Z}/2 \times \mathbb{Z}/2$ -action by row/column parity, and the row-based decomposition is invariant only under the diagonal subgroup. The 3×3 block in Case II is a small piece of *off-diagonal* symmetry that handles the remaining cases.